
PHYSICS

9702/22

Paper 2 AS Level Structured Questions

March 2019

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **9** printed pages.



Cambridge Assessment
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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Question	Answer	Marks
1(a)	kilogram / kg	B1
	kelvin / K	B1
1(b)	units for v : m s^{-1} <u>and</u> units for F : kg m s^{-2}	C1
	units for e : A s	C1
	units for μ : $\text{m s}^{-1} \text{ A s} / \text{kg m s}^{-2}$ $= \text{A kg}^{-1} \text{ s}^2$	A1

Question	Answer	Marks
2(a)(i)	distance in a specified direction (from a point)	B1
2(a)(ii)	change in velocity / time (taken)	B1
2(b)(i)	constant velocity so no resultant force	B1
	no resultant force so in equilibrium	B1
2(b)(ii)	(difference in height =) $47 \times 2.8 \times 60 \times \sin 24^\circ = 3200 \text{ m}$	A1

Question	Answer	Marks
2(b)(iii)	1 $(\Delta)E = mg(\Delta)h$ $= 85 \times 9.81 \times 3200$	C1
	$= 2.7 \times 10^6 \text{ J}$	A1
	2 <u>In terms of energy:</u> work done $= 2.7 \times 10^6 \text{ J}$ force $= 2.7 \times 10^6 / (47 \times 2.8 \times 60)$	C1
	$= 340 \text{ N}$	A1
	<u>In terms of forces:</u> component of weight along path = force due to air resistance force $= 85 \times 9.81 \times \sin 24^\circ$	(C1)
	$= 340 \text{ N}$	(A1)
2(b)(iv)	$(\Delta)p = \rho g(\Delta)h$ $(92 - 63) \times 10^3 = \rho \times 9.81 \times 3200$	C1
	$\rho = 0.92 \text{ kg m}^{-3}$	A1

Question	Answer	Marks
3(a)	$(m \times 3.0) \text{ or } (2.5 \times 9.6 \times \cos 60^\circ)$	C1
	$(m \times 3.0) - (2.5 \times 9.6 \times \cos 60^\circ) = 0 \text{ so } m = 4.0 \text{ (kg)}$	A1

Question	Answer	Marks
3(b)	$2.5 \times 9.6 \times \sin 60^\circ = (4.0 + 2.5) \times V$	C1
	$V = 3.2 \text{ m s}^{-1}$	A1
	or use of momentum vector triangle: $(4.0 \times 3.0)^2 + [(4.0 + 2.5) \times V]^2 = (2.5 \times 9.6)^2$	(C1)
	$V = 3.2 \text{ m s}^{-1}$	(A1)
3(c)	$E = \frac{1}{2}mv^2$ difference in $E_K = \frac{1}{2} \times 2.5 \times (9.6)^2 - \frac{1}{2} \times 4.0 \times (3.0)^2$	C1
	$= 97 \text{ J}$	A1

Question	Answer	Marks
4(a)	force per unit positive charge	B1
4(b)(i)	1 $E = V/d$ or $E = \Delta V / \Delta d$ $d = 4.0 \times 10^3 / 5.0 \times 10^4$	C1
	$= 8.0 \times 10^{-2} \text{ m}$	A1
	2 plates are (in) horizontal (plane) (above and below the rod)	B1
	top (plate) negative and bottom (plate) positive	B1
4(b)(ii)	magnitude $= 5.0 \times 10^4 \times 3 \times 1.6 \times 10^{-19}$ $= 2.4 \times 10^{-14} \text{ N}$	A1
	direction is (vertically) downwards / down	B1

Question	Answer	Marks
4(b)(iii)	$6.2 \times 10^{-16} = 2.4 \times 10^{-14} \times 72 \times 10^{-3} \times \cos \theta$	C1
	$\theta = 69^\circ$	A1

Question	Answer	Marks
5(a)(i)	(two) waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of the displacement of each wave	B1
5(a)(ii)	constant phase difference (between the waves)	B1
5(b)	$I \propto A^2$	C1
	$3I / I = (A + 1.5)^2 / 1.5^2$	
	$A = 1.1 \text{ cm}$	A1
5(c)(i)	$\lambda = ax / D$	C1
	e.g. $a = 680 \times 10^{-9} \times 2.0 / 4.0 \times 10^{-3}$	C1
	$a = 3.4 \times 10^{-4} \text{ m}$	A1
5(c)(ii)	straight line from positive value on x-axis and always below 'old' line	B1
	straight line with a smaller positive gradient than 'old' line	B1

Question	Answer	Marks
6(a)	e.m.f.: energy transferred from chemical to electrical (per unit charge)	B1
	p.d.: energy transferred from electrical to thermal (per unit charge)	B1

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Question	Answer	Marks
6(b)(i)	1 $I = 4.8 / 32$ $= 0.15 \text{ A}$	A1
	2 $P = EI$ or $P = VI$ or $P = I^2 R$ or $P = V^2 / R$ $= 6.0 \times 0.15$ or $0.15^2 \times 40$ or $6.0^2 / 40$	C1
	$= 0.90 \text{ W}$	A1
	3 number $= It / e$ $= [0.15 \times 25] / 1.6 \times 10^{-19}$	C1
	$= 2.3 \times 10^{19}$	A1
	or $Q = 0.15 \times 25 (= 3.75)$ number $= 3.75 / 1.6 \times 10^{-19}$	(C1)
	$= 2.3 \times 10^{19}$	(A1)
	4 $4.8 / 6.0 = 32 / (R_{XY} + 32)$ or $1.2 / 6.0 = R_{XY} / (R_{XY} + 32)$ or $4.8 / 1.2 = 32 / R_{XY}$	C1
	$R_{XY} = 8.0 \Omega$	A1
	Alternative methods:	
	$R_{XY} = (6.0 - 4.8) / 0.15$ or	(C1)
	$= 8.0 \Omega$	(A1)
	or $6.0 = 0.15 (32 + R_{XY})$	(C1)
	$R_{XY} = 40 - 32$ $= 8.0 \Omega$	(A1)

Question	Answer	Marks
6(b)(i)	5 $1/8.0 = 1/R_x + 1/24$	C1
	$R_x = 12\ \Omega$	A1
	Alternative method:	
	$I_Z = 4.8/32 = 0.15$ and $I_Y = 1.2/24 = 0.05$ $I_x = 0.15 - 0.05 (= 0.10)$	(C1)
	$R_x = 1.2/0.10 = 12\ \Omega$	(A1)
6(b)(ii)	<u>total</u> resistance decreases	M1
	(so voltmeter) reading increases	A1

Question	Answer	Marks
7(a)(i)	alpha, neutron and proton	B1
7(a)(ii)	neutron	B1
7(a)(iii)	beta plus or β^+	B1
7(b)	$\bar{d}$ has charge $(+)\frac{1}{3}e$	C1
	(so) other quark has charge $= e - \frac{1}{3}e$ $= (+)\frac{2}{3}e$	M1
	other quark is an up / u	A1